TRANSFINITTE

PROBLEM STATEMENTS

MANAGEMENT

McKinsey & Company

Reward worth Rs.40k for the winning team (Only 1 winner)

PS1

Case: The Dragon Passion Problem at Pulp Riot Juice Co.

Background

Pulp Riot Juice Co. is a fast-growing Indian maker of bottled fruit juices, known for unusual flavors and highprofile marketing. It operates two plants and one co-packer, and ships to nine depots that serve about 600 general-trade distributors, modern-trade retailers and quick-commerce dark stores across the country. Annual revenue is ₹1,200 crore, at a contribution margin of 32%. Its range has five flavors: 1. Mango Peach Basil (38% of summer volume) uses mango pulp from contracted processors, peach pulp from hill-state orchards and fresh basil from contract farms near the plant, pressed into extract on site. 2. Guava Chilli (20%) uses guava pulp from two regional suppliers. 3. Passionfruit Punch (16%) depends on a single hill-region cooperative. 4. Dragonfruit Bloom (14%) uses dragonfruit pulp that is 60% domestic and 40% imported, with a lead time of six to eight weeks. 5. Dragon Passion (12%), a new blend, uses both dragonfruit and passionfruit pulp. Peak summers (March to June) bring in 48% of annual revenue. The plants cannot add mid-season lines. The copacker can add up to 10% volume at a premium.

Key operating data:

• **Sales mix:** General trade accounted for 55% of summer volume, modern trade 25% and quick commerce 20%. The average realized price was ₹60 per bottle. • Planning: Forecasts are updated monthly, and forecast error averaged 30% at the flavor and depot level. Quick-commerce platforms send daily sell-through reports to marketing; planners receive consolidated sales data at month-end.

• **Production:** Summer volume was about 9.6 crore bottles. The two plants can make about 2.5 crore bottles a month and ran at 95% utilization through the summer. The co-packer's extra volume (up to 10%) costs 25% more per bottle than in-house production and needs two weeks' notice.

• **Sourcing:** Procurement targets are set on price per kilogram, and no second passionfruit supplier has been qualified. Imported dragonfruit pulp is ordered once a quarter. Marketing set Dragon Passion's launch date and informed procurement two weeks before launch.

• **Shelf life:** Frozen fruit pulp keeps for up to six months in plant cold storage. Bottled juice has a 90-day shelf life, and modern-trade and quick-commerce buyers reject stock with less than 60 days left. • Routes: Passionfruit pulp normally reaches the plants by road in three days. There is no alternative road route from the cooperative, and the highway has closed for at least a week in three of the last five summers. The plants held about two weeks of passionfruit pulp when the May closure began, and air freight was booked in its second week.

• **Depots:** The nine depots sit in four regions. Each depot holds about three weeks of cover for every flavor, based on the previous year's flavor mix, and allocations are reset monthly. Each depot reports its stock weekly in its own spreadsheet. A transfer between depots takes two to four days and costs about ₹1.50 per bottle.

In early March, Pulp Riot launched Dragon Passion. Two weeks after launch, an influencer's 30-second reel featuring the drink went viral, and quick-commerce orders rose fourfold over the next two weeks. Planners saw the spike only when distributors and retailers began reporting shortages. Then, in May, landslides closed the only highway from the passionfruit cooperative for three weeks. The company spent ₹3.2 crore air-freighting pulp at five times the normal cost, cut Passionfruit Punch production to protect Dragon Passion, and still ran out of both products. Meanwhile, some depots were overstocked with Mango Peach Basil and Guava Chilli, which later cost ₹6 crore in write-offs and discounts, while other regions ran short of the same flavors. About 8% of summer revenue was lost to stock-outs.

Within six weeks of the stock-outs, a larger rival launched its own dragonfruit-passionfruit drink and took the shelf space Pulp Riot couldn't fill.

The CEO’s verdict: “We didn’t lose to the competitor; we lost to ourselves.”

Problem Statement

Pulp Riot’s planning involves marketing, sales, procurement, supply chain and finance, and is largely manual.
Key challenges include:
• Demand changes were not reflected in plans until shortages appeared.
• Passionfruit pulp ran short during the peak, and summer plant capacity is fixed, with the co-packer adding only up to 10% at a premium.
• Some flavors and regions ran out while stock built up elsewhere, costing ₹6 crore in write-offs and discounts.
• Stock did not reach the depots and channels where demand was rising, and there is no general-trade outlet data.
• The May highway closure cut off passionfruit pulp for three weeks, and imported dragonfruit pulp takes six to eight weeks. Air freight cost ₹3.2 crore at five times the normal cost. Peak-season trade-offs are escalated to the COO, and the S&OP meeting is held monthly. Fifteen planners work in spreadsheets, there is no data or AI team, IT is run by vendors, and two earlier digital pilots stalled after go-live.

Task

Build a model or agent that gives Pulp Riot end-to-end visibility across demand, supply, inventory, distribution and transportation, identifies risks early and recommends how to mitigate them, so that last summer’s failures do not recur next summer, eight months away. The model can be built in any tool, but the full rollout of your solution must fit within a total first-year budget of ₹2.5 crore, covering all one-time build costs and first-year running costs, with vendor-run IT, no general-trade outlet data, fixed plant capacity in summer, and a stakeholder approving every recommendation it makes. The model should cover all five areas at a high level, with depth on the two modules you choose under Objectives. No dataset is provided: build sample data from the figures in this case, which contains all the information you need. The focus is on how you identify the problems and reason through solving them, not on a polished system design.

Note: You may use vibe-coding tools or AI agent builders to create the model or agent.

Objectives

Choose exactly two of the modules below and show how they are linked: how a problem in one module caused or worsened problems in the other, and vice versa. For those two modules, identify the problems from the case facts, trace them to their root causes, and show how the model or agent would detect and mitigate the related risks across both. The solution must be resilient to disruptions, scalable to more flavors, depots and channels, integrated across the five areas, and efficient in cost and effort. Base each answer on the facts in this case; it contains all the information needed, so no outside assumptions are required. The case does not prescribe a solution. • Demand: What went wrong in demand planning last summer, and how should the model catch the next shift in demand early?
• Supply: What left supply exposed last summer, how large is the risk for next summer, and how should the model flag and respond to it?
• Inventory: Why did some flavors stock out while others were written off, and how should the model prevent both next summer?
• **Distribution:** Why did stock end up in the wrong places, and how should the model get it to where demand is?
• Transportation: What made inbound logistics fragile last summer, what did it cost, and how should the model reduce that exposure?

Deliverables Expectation

• Problem identification: The key problems in your two chosen modules, traced to their root causes, showing the cause-and-impact link in both directions: how problems in the first module affected the second, and how problems in the second affected the first, with last summer's loss quantified and attributed to those problems.
• Working model or agent: A working prototype (spreadsheet, script or low-code tool) that gives an end-toend view, with depth on your two chosen modules and the links between them, and flags risks with recommended actions. Show how it is resilient, scalable, integrated and efficient.
• Scenario test: A replay of last summer's viral spike and highway closure, as they affected your two chosen modules, showing what the model would have

flagged, when, and what it would have recommended. • Data and logic: The case data and decision rules behind the model, the trade-offs you made, and where a human makes the final call.
• Impact and next steps: The expected impact on next summer within the ₹2.5 crore budget, and what you would test or build next.

Submission

Teams of up to four have 42 hours to submit a deck of no more than 10 slides, including opening and closing slides, and the working prototype. Each team then presents for 10 minutes, followed by 5 minutes of questions from the judges.

Evaluation Criteria

Dimension	Weight	Description
Model or Agent Build	25%	How well the model gives end-to-end visibility, flags risks early and supports mitigation decisions.
Strategic Understanding	30%	Clarity in identifying the business problems and tracing them to their root causes.
Creativity	20%	Original, practical ideas for how the model detects risks and supports decisions.
Feasibility	15%	Use of the case data, and realism of timeline, capabilities, adoption and budget.
Presentation Clarity	10%	A clear, logical storyline from problem to solution.

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Inqor GTM Hackathon Student Brief

1. About Inqor

Inqor.ai is an AI research platform that runs in-depth qualitative interviews with real people at scale and turns them into cited, board-ready findings in 48–72 hours. It was built at NIT Trichy. The product is now live, and this problem statement is about taking it to market.

Born at TransfiNITTe 2026

Inqor came out of last year's TransfiNITTe, NIT Trichy's hackathon. The first set of Inqor developers were identified at that event, and the codebase was built from the work of the top 3 teams. What started as hackathon code is now a complete, live SaaS product with self-serve sign-up, billing and region-based pricing.

Inqor is backed by Dare 2 Dream Ventures, an angel fund started by NIT Trichy alumni to support ventures by students and alumni. Last year, students built the product. This year, we want students to build the engine that sells it.

The problem Inqor solves

Companies that want to understand their customers, employees or market have had two options, and both fall short:

- **Surveys scale to thousands of people but stay shallow. A score tells you *what* happened, never *why*.**
- **In-depth interviews get to the why, but they need trained moderators, take 1–2 weeks and cost a lot, so teams can only afford a handful.**

Inqor gives the depth of an interview at the scale of a survey. Hundreds of people can be interviewed in parallel, and every insight is traced back to the interview it came from.

How it works

1. **Brief. The customer defines the research objective, target audience and key questions, and can upload supporting material.**
2. **Orchestrate. Inqor builds the discussion guide and configures the study, including audience criteria, interview logic and timelines.**
3. **Interview. Respondents are invited by email. Sophia, Inqor's AI interviewer, holds a one-on-one conversation with each of them and asks follow-up questions in real time when an answer is vague or interesting.**
4. **Insight. All interviews are synthesised into themes and cited findings, delivered as a report and a slide deck.**

A 2-minute walkthrough video is on the Inqor website.

The three use cases

Use case	Typical buyer	The question it answers	Moments that trigger a purchase
Customer Satisfaction (CSAT)	Heads of customer success or experience, founders, product leaders	Why do customers rate us the way they do, and what would move the number?	CSAT or NPS drop, rising churn, post-onboarding check-in, pre-renewal review
Employee Satisfaction	HR and people leaders, founders of growing teams	What is really driving engagement, morale or attrition?	Annual engagement survey results, attrition spike, post-reorg or post-merger pulse, stay and exit interviews
Market Research	Product managers, marketers, founders, consultancies, research agencies	What do buyers need, and how do they react to our idea, product or price?	New product or feature concept, entering a new market, pricing change, competitive loss analysis

Pricing and free trial

Inqor charges for completed interviews, not for software seats. Discussion guides never use credits.

Plan	Price (India)	What you get
Free trial	Free, no credit card	5 credits, valid for 7 days, up to 3 discussion guides, AI-led interviews with follow-ups
Subscription	₹4,000 per month + GST	10 credits every month, unlimited discussion guides, reports and slide decks; each month's credits stay valid for 12 months
Add-on pack	₹5,000 per pack + GST	25 credits per pack, valid for 12 months, one-time purchase, buy as many as needed
Enterprise	Custom	Volume plans for large teams or multiple markets

Inqor charges for completed interviews, not for software seats. Discussion guides never use credits.

How credits work:1 credit = 1 respondent completing an interview of up to 5 questions (typically up to 15 minutes).

- Longer guides use more credits in steps of 5 questions: 6–10 questions = 2 credits, 11–15 = 3, 16–20 = 4 per respondent.
- A credit is used only when the respondent finishes. Drop-offs cost nothing.
- Customers in India see INR pricing plus GST. Customers elsewhere see USD pricing at checkout.

Full details are on the pricing section of the website, including a calculator that shows how many credits a study needs.

The website today

www.inqor.ai is currently a single page with these sections: How It Works, a 2-minute demo video, Solutions, a comparison against traditional interviews, Pricing with the credit calculator, and a "Start Free" call to action. The Solutions section currently features Customer Satisfaction and Employee Satisfaction. Market Research does not yet have its own content. Treat the site as your starting point, not a finished asset.

Where Inqor stands now

- Product: complete and live, with self-serve sign-up, a free trial and online payments.
- Market: anyone in the English-speaking world who wants to understand their customers, employees or market. Teams may choose which geographies and segments to prioritise, as long as they justify the choice.
- Visibility: close to zero. Search for Inqor today and you will find almost nothing: no reviews, no content, no backlinks, and no mentions in answers from ChatGPT, Perplexity or Gemini.

That last point is the opportunity. Inqor has a working product and almost no market presence. Your plan decides how it gets found, tried and bought.

2. The challenge

Build a complete, executable go-to-market plan that turns Inqor from an unknown product into one that real customers find, try and pay for. This is not a coding hackathon. Do not build new features for Inqor. Your work is on how Inqor reaches its market.

The winning plans will not just be presented. The top 3 teams will get real money to run them, and they will earn commission on the customers they bring in. Write a plan you would bet your own budget on, because you will.

We are looking for plans that are:

- Specific. Name the keywords, audiences, accounts, pages and messages. Avoid generic "use

social media" advice.

- Measurable. Every activity links to a number: impressions, clicks, trial sign-ups, paid conversions, cost per acquisition.
- Budget-realistic. The first phase runs on ₹10,000 of marketing spend. Show what that buys and what you expect back.
- Executable by your team. Plan only what your team can actually do in the execution window, with the tools and access available.

3. What your GTM plan must cover

Every plan must have a strategic foundation and all five channel workstreams below. You can weight the channels differently, but you must justify where you put your effort and money.

Foundation: who, what and how you will measure it

- Ideal customer profile (ICP). Which use case, which company size, which industries, which geographies and which job titles you will target first, and why.
- Positioning and messaging. A one-line positioning statement, the core message per use case, and how Inqor compares to surveys (Typeform, SurveyMonkey, Google Forms), employee experience platforms (Culture Amp, Qualtrics EX), research agencies and other AI interview tools.
- Funnel and metrics. Define the funnel from first visit to free trial to first paid subscription. Set targets for each stage and explain how you will track them (UTM parameters, Google Analytics 4, ad platform pixels).

Workstream A: SEO (search engine optimisation)

Goal: Inqor ranks on Google for the searches its buyers actually make.

- Keyword research: 30 or more target keywords grouped by use case and by intent (problem-aware, solution-aware, comparison), with estimated volume and difficulty.
- Technical SEO audit of inqor.ai: page speed, mobile, indexing, metadata, structured data and site architecture.
- A proposed site structure, for example dedicated pages per use case, per industry and "Inqor vs" comparison pages.
- A content calendar of at least 10 pieces (blogs, guides, templates) with target keyword, outline and publishing date. Include 2 fully written sample pieces.
- A backlink and off-page plan: directories (G2, Capterra, Product Hunt, SaaS listings), guest posts, partnerships and communities.

Workstream B: LLM optimisation (GEO / AEO)

Goal: when someone asks ChatGPT, Perplexity, Gemini or Claude for the best way to understand customers or employees, Inqor appears in the answer.

- **A baseline audit: 20 or more prompts a buyer might ask an AI assistant, and what each assistant answers today.**
- **The sources these assistants cite in Inqor's category (review sites, Reddit, listicles, comparison articles) and a plan to earn a presence in them.**
- **On-site changes that make Inqor easy for AI to read and quote: clear factual pages, FAQs, structured data, consistent descriptions of what Inqor is.**
- **A way to track AI visibility over time (repeatable prompt set, share-of-answer tracking).**

Workstream C: Website optimisation

Goal: more of the people who land on inqor.ai start a free trial.

- **A conversion audit of the current site: messaging clarity, page flow, calls to action, trust signals, mobile experience and load speed.**
- **Recommended changes, prioritised by expected impact and effort. Include mock-ups or rewritten copy for at least the hero section and one use-case page.**
- **Landing page plans for your paid campaigns, matched to each campaign's audience and message.**
- **An A/B testing plan: what you will test first, the hypothesis and the success metric.**
- **Trial activation ideas: how to get trial users to run their first study within the 7-day window.**

Workstream D: PPC campaigns (paid search and paid social)

Goal: paid traffic that converts into trials and paid customers at a cost Inqor can sustain.

- **Channel choice and split: Google Search, LinkedIn, Meta, Reddit or others, with reasons.**
- **Campaign structure: campaigns, ad groups, keywords with match types, negative keywords and audience targeting (job titles, industries, company sizes).**
- **Ad copy: at least 3 variants per campaign, plus creative concepts for social ads.**
- **Budget allocation of the ₹10,000, bidding approach, daily caps and a week-by-week plan.**
- **Expected cost per click, click-through rate, conversion rate and cost per trial and per paid customer, with your assumptions shown.**
- **Rules for when to pause, cut or scale a campaign.**

Workstream E: Outbound email and account-based marketing (ABM)

Goal: start conversations directly with the companies most likely to buy.

- **Target account list: a sample of at least 50 named companies that fit your ICP, with the reason each one fits.**

- **Buyer personas and how you will find their contact details (LinkedIn, Apollo, company websites), using only legitimate, permission-respecting sources.**
- **Email sequences: at least one 3–4 step sequence per priority use case, with subject lines and full copy.**
- **A personalisation approach for top accounts, such as trigger-based outreach on a new funding round, a leadership hire or bad public reviews.**
- **Supporting touches: LinkedIn outreach, content offers and a free pilot study offer.**
- **Deliverability and compliance: domain warm-up, sending limits, unsubscribe handling, and compliance with anti-spam rules such as CAN-SPAM, GDPR and India's DPDP Act.**

4. Deliverables

Each team submits three items. Plans that skip a workstream or leave out numbers will be marked down.

Deliverable	Format	Limit	What it must contain
GTM plan document	PDF	Up to 25 pages, plus appendices	Foundation, all five workstreams, a 30-day execution calendar, budget allocation, target metrics and risks
Pitch deck	PDF or PPT	Up to 15 slides	The plan's core logic, budget, expected results and why your team will execute it best
Execution sheet	Google Sheet or Excel	No limit	Week-by-week activities, owners in your team, spend per channel and target versus actual tracking columns

Appendices can include keyword lists, the target account list, full ad copy, email sequences, website mock-ups and the AI-visibility prompt audit.

Finalists will present live: 10 minutes of pitch followed by 10 minutes of questions from the jury.

5. Rewards

The top 3 teams win cash prizes and a ₹10,000 marketing budget each to execute their plan, and they earn 5% commission on every customer they bring in.

Rank	Prize money	Marketing spend budget	Commission
1st	₹15,000	₹10,000	5% of each attributed customer's first payment
2nd	₹10,000	₹10,000	5% of each attributed customer's first payment
3rd	₹5,000	₹10,000	5% of each attributed customer's first payment

Marketing spend budget

- **The ₹10,000 is for executing the plan only: ad spend and essential tools. It is not prize money and cannot be paid out as cash.**
- **Spend must follow the budget in the team's winning plan. Changes of more than 20% between channels need approval from the Inqor team.**
- **Every rupee spent must be reported in the execution sheet with receipts or platform invoices.**

Commission

- **Teams earn 5% of the first payment Inqor receives from each customer they bring in. This is a one-time commission per customer, not a share of recurring or annual revenue.**
- **Commission is calculated on the amount before GST.**
- **Example: a customer who subscribes at ₹4,000 and buys one ₹5,000 add-on pack in the same first payment earns the team 5% of ₹9,000 = ₹450.**
- **Customers are attributed to a team through the team's unique tracking links and referral or coupon codes. A customer counts only if their sign-up is recorded against the team's tracking.**
- **Free trial sign-ups do not earn commission on their own. The commission is earned when that trial converts into a first payment.**

6. How the top 3 teams will be selected

The jury scores every plan out of 100. Plans are rewarded for being measurable and executable, not for polish. The 3 highest scores win.

Criterion	Weight	What the jury looks for
ICP, positioning and messaging	15	A sharp, well-reasoned choice of first customers and a message that clearly separates Inqor from surveys, agencies and other AI tools
Channel depth across all five workstreams	30	Specific, expert-level work in SEO, LLM optimisation, website, PPC and outbound/ABM: real keywords, real accounts, real copy and real page changes (6 points per workstream)
Measurability and funnel economics	20	Clear targets at every funnel stage, realistic assumptions, and believable cost per trial and cost per paying customer
Budget realism and execution plan	15	A sensible split of the ₹10,000, a week-by-week calendar, clear owners in the team, and rules for cutting or scaling spend
Creativity and insight	10	Fresh ideas that a typical agency would miss, and evidence that the team actually used Inqor and studied the market
Pitch and Q&A	10	Clear story, command of the numbers and convincing answers under questioning
Total	**100**	

Bonus points (up to 5)

- **Running a real Inqor study during the hackathon using the free trial, for example interviewing potential buyers to test your messaging, and using the findings in your plan.**
- **Early evidence of traction gathered legitimately, such as interest from a named target account.**

Tie-breaker

If two teams have the same score, the team with the higher score for measurability and funnel economics ranks higher.

Disqualification

- **Plagiarised or wholly AI-generated plans submitted without the team's own analysis. AI tools may be used to assist, but the thinking must be the team's.**

- Any plan that relies on spam, bought email lists, fake reviews, misleading claims about Inqor, or other unethical or illegal tactics.
- Missing any of the three required deliverables.

7. Execution phase for the winning teams

After the results, the top 3 teams run their plans with real money for a fixed execution window. The Inqor team will support them and protect the brand.

What the Inqor team provides

- Access to ad accounts set up and owned by Inqor, with team members added as users.
- Each team's unique tracking links and referral or coupon codes for attribution.
- Brand assets: logo, colours, product screenshots and the demo video.
- A sending domain or mailbox for outbound email, set up by Inqor.
- A shared dashboard or weekly report on trials and paid sign-ups attributed to each team.

Rules for the teams

- All ads, landing pages, emails and public content must be approved by the Inqor team before going live.
- Changes to the live inqor.ai website are made through the Inqor team, based on the team's recommendations.
- Teams must not make claims about Inqor's features, customers, results or security that the Inqor team has not confirmed.
- Teams submit a short weekly update: spend, results against target and what they will change next week.
- A final results presentation closes the execution window.

Approach to compliance

Outbound email must use legitimately sourced business contacts, include a clear unsubscribe option and respect the anti-spam and data protection rules of the recipient's country. Teams must never send from personal accounts in Inqor's name.

8. Eligibility, timeline and FAQ

Eligibility

- Open to current NIT Trichy students. Team size and year or department eligibility: *to be confirmed*.

- **Each student may be part of only one team.**

Timeline

Milestone	Date
Hackathon brief released and registrations open	To be confirmed
Product walkthrough and Q&A with the Inqor team	To be confirmed
Plan submission deadline	To be confirmed
Finalist presentations and results	To be confirmed
Execution window for the top 3 teams	To be confirmed

Resources

- **Website and pricing: www.inqor.ai**
- **Product: every team can sign up for the free trial (5 credits, 7 days, no credit card) to experience Inqor first-hand.**
- **Questions: the Inqor team will hold an open Q&A session after the brief is released. Contact details: *to be confirmed*.**

FAQ

Can we change the product or write code for Inqor? No. This hackathon is about go-to-market only. Product ideas can go in an appendix, but they will not be scored.

Do we have to cover all five workstreams? Yes. You can put more effort and budget into some channels than others, but every workstream needs a concrete plan.

Can we target markets outside India? Yes. Inqor's market is the entire English-speaking world. Choose your focus and explain why.

Who owns the plans we submit? Teams may be credited for their work. By submitting, teams allow Dare 2 Dream Ventures and Inqor to use ideas from any submitted plan.

How do we know which customers count for our commission? Only customers recorded against your team's unique tracking links or codes count. Share your links and codes everywhere you promote Inqor.

TRANSFINITTE

CONVERRA

PROBLEM STATEMENTS

CONVERRA GROWTH PARTNERS

ABOUT CONVERRA GROWTH PARTNERS

Who We Are

Converra Growth Partners is a sales enablement and revenue growth partner working with businesses across SaaS, IT services, EdTech and technology sectors.

We help businesses unlock revenue opportunities through lead generation, pre-sales, customer engagement, sales execution and deal closures. Our approach enables companies to access sales capabilities through a flexible, performance-driven partnership rather than having to build an entire sales department from scratch.

We aspire to become an additional revenue engine for technology companies, helping them turn their products and services into commercial opportunities.

Our Vision

To build a scalable, technology-enabled affiliate sales agency headquartered in Trichy, serving businesses across India and international markets through a structured, high-performing sales ecosystem.

Our vision extends beyond providing sales services. We aim to build an organisation that connects emerging technology companies with the sales talent, commercial partnerships and execution capabilities required to grow.

Why This Matters

The technology ecosystem continues to create opportunities for new SaaS products, AI-powered solutions, IT services and digital businesses. However, developing a product and building a revenuegenerating business are two different challenges.

Many emerging companies invest heavily in product development but face difficulties establishing a predictable sales pipeline, reaching the right customers and converting market opportunities into revenue. Building an in-house sales and marketing team can also require considerable investment in recruitment, training, technology and management.

Converra aims to bridge this gap by providing businesses with a flexible route to sales execution and revenue generation.

The Opportunity

Can a Trichy-based company build a sales organisation that helps technology businesses acquire customers, generate demand for their products and expand into new markets without having to build every sales capability in-house?

We invite the next generation of problem-solvers to explore how Converra can develop a sustainable growth model that creates value for client businesses while building meaningful career opportunities for the people who power its growth.

From a Lean Freelance Model to a Scalable Organisation Converra has primarily operated through a lean, freelance-based model, with a small team supporting different stages of the sales cycle, including lead generation, pre-sales, demo booking, customer engagement and deal closure.

Revenue is generated through commercial partnerships and commissions earned from successful business conversions.

This operating model has provided a foundation for delivering sales support without the overhead of a large permanent workforce. The next stage is to explore how this foundation can evolve into a structured, scalable and commercially sustainable organisation.

Our Next Phase of Growth:

Converra's long-term ambition is to establish a stronger operating presence in Trichy, supported by an office-based team and a distributed remote workforce.

The goal is to build an organisation capable of supporting multiple client

businesses, expanding its sales capacity and serving customers across domestic and international markets, Mainly the United States & Singapore.

This ambition requires more than increasing the number of salespeople. It requires a deeper understanding of where commercial opportunities exist, how clients can be acquired, how demand can be generated for their products and how the organisation can expand without compromising financial sustainability.

The Strategic Challenges:

1. **Client Acquisition and Market Access** How can Converra consistently identify, reach and convert suitable technology companies into longterm revenue partnerships? The challenge is to determine which customer segments offer the strongest opportunities, how decision-makers can be accessed and how a repeatable pipeline of potential clients can be established.

2. **Lead Generation for Client Businesses** Acquiring a client is only the beginning. Converra must also help generate qualified sales opportunities for that client's products and services. Many SaaS, IT and technology businesses depend on inbound enquiries, existing networks or limited marketing channels. Some have strong products but lack an established demand-generation system. Generating leads for a product that Converra does not own presents a distinct challenge. The company must understand the client's target audience, product positioning, market awareness and sales cycle before determining how demand can be generated. How can Converra build a repeatable and commercially viable lead-generation capability across different client businesses?

3. **International Market Entry** Converra aims to explore opportunities beyond India, particularly in the United States and other international technology markets. Understanding market demand, customer expectations, competitive dynamics, acquisition channels and commercial requirements will be critical to determining where and how the company should expand.

4. **Revenue Sustainability** A commission-led business depends on more than

closing individual deals. Its sustainability requires a consistent supply of suitable clients, sufficient demand for the products being sold, effective lead conversion and commercial arrangements that support continued investment in the business.

5. **Workforce Development and Scalability** How can Converra build and retain a capable sales and marketing workforce in Trichy while balancing office-based employees with remote talent? The challenge is to create an environment where people contribute to measurable business outcomes while developing commercially valuable skills.

6. **Competitive Differentiation** Converra operates in a landscape that includes lead-generation agencies, performance marketing firms, sales outsourcing providers, independent consultants, affiliate networks and conventional inhouse sales teams. How can Converra establish a distinctive position by combining demand generation, sales execution and revenue partnerships in a way that creates measurable value for its clients? The central challenge is to determine how these elements can work together to build a repeatable growth engine rather than a business dependent on individual relationships or isolated sales successes.

THE CORE PROBLEM STATEMENT

The Challenge

How can Converra Growth Partners build a repeatable and commercially sustainable growth model that enables a Trichy-based sales agency to acquire technology clients, generate qualified demand for their products and services, enter international markets and scale its revenue and workforce over time?

Participants are expected to investigate the commercial opportunities available to Converra and develop an evidence-based strategy that addresses its market access, demand generation, competitive positioning and expansion ambitions.

The objective is not to redesign basic sales processes in isolation. It is to examine the strategic decisions that determine where Converra competes, how it acquires clients, how it generates demand for their offerings and how it converts those opportunities into sustainable revenue.

Strategic Questions to Explore

A. **Market Selection and Opportunity Identification:-** Which industries, business segments and customer profiles offer the strongest opportunities for Converra's sales partnership model?- Which technology businesses are most likely to benefit from an external sales and revenue growth partner?- How should market attractiveness, competitive intensity, sales-cycle duration and revenue potential influence market prioritisation?- What evidence should support the selection of target industries and customer segments?

B. Client Acquisition and Commercial Positioning:- How can Converra identify and access decision-makers in target companies?- Which acquisition channels and partnership routes offer promising opportunities for reaching potential clients?- What barriers might prevent a company from trusting an external partner with its sales function?- How can Converra demonstrate its value and differentiate itself from other sales service providers?- What would a repeatable client-acquisition engine look like, and how could its effectiveness be measured?

C. Lead Generation and Demand Creation:- How can Converra generate qualified leads for client products when inbound demand is limited or inconsistent?- How should the company evaluate the suitability of outbound prospecting, performance marketing, Meta advertising, content-led acquisition, strategic partnerships and other potential channels?- How can the appropriate channel be determined for products with different audiences, price points, market awareness and sales cycles?- How can Converra distinguish between generating enquiries and generating qualified opportunities that can convert into revenue?- What framework could help evaluate lead quality, acquisition costs, conversion potential and commercial returns across multiple client businesses?

D. US and International Expansion:

- What opportunities might exist for a Trichy-based sales agency in the United States and other international technology markets?

- Which customer segments and geographic markets should be evaluated first?

- How should Converra assess market-entry costs, customer expectations, competition and commercial requirements?

- What evidence would justify entering a market, and how should the company assess whether its expansion is commercially viable?

E. Scaling the Business-

How can client acquisition, demand generation, sales productivity and workforce expansion be aligned?

- What commercial and financial conditions should support decisions about hiring and infrastructure investment?

- How can Converra expand its capacity while protecting service quality, client relationships and profitability?

- What would a sustainable growth model look like over the next 12 months and beyond?

Participants may investigate additional questions where their research identifies meaningful opportunities, risks or alternative growth pathways.

RESEARCH SCOPE AND EXPECTED DELIVERABLES:

Turn Business Questions into an Actionable Growth Strategy Teams are expected to develop a practical business blueprint supported by market research, commercial logic, realistic assumptions and a clear implementation plan.

Research Track 1: Market and Customer Intelligence

Research Track 2: Client Acquisition Strategy

Research Track 3: Demand Generation and Lead Acquisition Research Track 4: International Market Expansion

Research Track 6: Growth Economics and Workforce Scaling Research Track 7: Execution Roadmap

Final Deliverables

1. Executive Summary: The business challenge, key findings and principal recommendation.

2. Market Research: Evidence supporting the selected customer segments and target markets.

3. Client Acquisition Strategy: A proposed approach to building a repeatable source of qualified client opportunities.

4. Lead Generation Strategy: A framework for selecting, testing and scaling acquisition channels for different client products.

5. International Expansion Assessment: A reasoned evaluation of the US market and other selected markets.

6. Competitive Analysis: Relevant alternatives, differentiation opportunities and positioning.

7. Commercial and Financial Model: Revenue scenarios, acquisition costs, operating expenses and break-even assumptions.

8. Implementation Roadmap: Prioritised actions, milestones, risks and KPIs.

9. Final Presentation: An evidence-led business pitch explaining the recommended strategy and its commercial viability.

Teams are encouraged to use credible public sources, transparent assumptions, market comparisons and visual frameworks to support their conclusions.

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THE HACKATHON Build.

Learn. Earn. Grow.

A Real-World Opportunity with Converra Growth Partners We believe that meaningful learning happens when knowledge is applied to real business challenges. Converra aims to provide selected students with an opportunity to continue beyond the hackathon through a structured, three-month or extended internship focused on business development, sales execution and revenue growth. This opportunity allows students to understand how a business identifies opportunities, engages customers, develops commercial relationships and converts market demand into revenue. What Students Can Gain

1. Hands-on Business Experience Work on live business opportunities and gain exposure to the practical challenges involved in developing and growing a business.

2. Performance-Linked Earning Opportunities Selected interns may contribute to eligible sales campaigns and earn incentives on successfully closed deals, subject to the applicable commercial and incentive terms.

3. Mentorship and Industry Exposure Learn from experienced sales professionals, business leaders and advisors, with exposure to different industries and commercial environments. 4. End-to-End Business Development Learning Develop practical understanding of prospecting, lead qualification, customer engagement, sales presentations, follow-ups, negotiations, CRM usage, pipeline management and deal closures.

5. Professional Recognition Eligible interns will receive an internship certificate upon meeting the applicable programme requirements, along with the opportunity to build practical experience in sales and business development.

6. Recognition for Exceptional Performance High-performing participants may be considered for additional rewards, gifts or hampers during the internship, based on performance and programme criteria.

7. Potential Career Progression Interns who demonstrate strong performance,

ownership and commercial capability may be considered for extended internships or future fixed stipend may be considered based on performance, role requirements and business needs.

Why This Is Different

The opportunity is designed to connect learning with real business outcomes.

Students can gain exposure to customer-facing work, understand how commercial decisions are made and see how individual contributions can influence revenue generation.

Beyond sales execution, participants may gain exposure to market research, lead-generation strategies, client acquisition, business development, account management, growth strategy and revenue operations.

The internship is intended to provide meaningful professional development, with transparent expectations around responsibilities, performance, incentives and any subsequent employment opportunities.

Our Closing message

Think strategically. Research deeply. Build for scale

Converra Growth Partners is at a stage where strategic thinking can translate directly into business action.

We invite students to challenge assumptions, investigate opportunities, question conventional approaches and develop solutions that can help a growing Trichy-based business compete in a wider market.

The challenge is not simply to grow a sales team. It is to discover how a local business can consistently acquire clients, generate demand for their products, convert opportunities into revenue and build sustainable growth across markets.

Building a Scalable, Technology-Enabled Revenue Growth Partner from Trichy.

Converra Growth Partners - Your Partner in Revenue Growth.

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HARDWARE



DARE TO DREAM

Hackathon Problem Statement: Early Warning of Worsening Heart Failure

Background

Problem Statement: How can we spot the early signs of worsening heart at home with almost no effort from the patient? Design a system that sends an alert several days before the patient needs hospital care. Your system has three parts. First is a sensing method. Second is a hardware design. Third is software that turns the data into an alert. An elderly patient in an Indian home should be able to use it every day.

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The Core Constraint: Effortless Capture

Your solution must follow all three rules:

Low effort: the patient does nothing or spends less than 2 minutes a day. A sticky patch worn all day may not be ideal

Real home use: patients may read little or speak a local language. Power cuts and weak internet are common. Your design must handle these.

Low cost: the parts for the home unit cost less than ₹5,000.

Your Creative Freedom

You choose what to sense. You choose where and how to sense it. You also choose how the data becomes an alert. These ideas are provided only to get you thinking on what's possible and are untested for heart failure. You can come up with a completely different idea

A smart scale that records a second signal during the daily weigh-in. One option is foot impedance. Impedance is how strongly the body resists a tiny electric current. Extra body fluid lowers it. Another option is the tiny body vibration made by each heartbeat.

A phone check-in. The camera can read the pulse from a fingertip. The microphone can listen to breathing or cough. A photo can show ankle swelling.

Sensors in the bed, chair or floor. They can track heart rate and breathing. They can also show how the patient sleeps.

Lying flat can make breathing hard. Data from a watch or ring the patient already owns.

A voice or text check-in in the local language. Combine it with sensor data.

Warning signs you can target are weight gain, a faster heart rate at rest, faster breathing, less activity, fluid in the lungs or legs, and trouble breathing when lying flat. One signal is rarely enough. Combining signals is encouraged.

What Teams Must Deliver

1. User journey: who uses it, when, and for how long? Why will they keep using it for months?

2. Hardware design: a block diagram (boxes showing how the parts connect) and a circuit diagram (called a schematic) for at least one custom hardware part. This part needs a sensor circuit and a microcontroller. A microcontroller is a small chip that runs code. Add a parts list with prices in INR. Building a physical device is optional.

3. Software and analytics: code that turns raw sensor signals into an alert. It learns each patient's own normal range. We call this the personal baseline. Each alert says why it fired. A simple screen shows the status to the patient, the family or the doctor.

4. Validation: test your system on public or computer-generated data. Report the results with the measures below. State the limits of your work.

Free Simulation Tools

Show proof of simulation from at least one free tool.

Tool	Use
Wokwi	Runs microcontroller code with virtual sensors in a browser
Tinkercad Circuits	Simple circuit simulation for beginners
LTspice	Designs the circuit that amplifies and cleans a sensor signal
KiCad	Draws circuit diagrams and simulates them
EasyEDA	Draws and simulates circuits online
Python (NeuroKit2, SciPy)	Creates fake body signals and tests signal processing

Data and Validation

The BIDMC Congestive Heart Failure Database holds long ECG recordings from 15 patients. An ECG is a recording of the heart's electrical signal. You can use it to test code that measures heart rate. No public data matches most home sensing ideas. So create computer-generated (synthetic) data for your own signals. Make at least 20 virtual patients with 30 days of data each. In each patient, make the heart failure worsen 3 to 10 days before a hospital visit. Explain how you built the data.

Report these measures: Sensitivity: the share of real worsening events your system catches.

Specificity: the share of normal days your system correctly leaves alone.

Lead time: how many days before the hospital visit the alert fires. Report the median (middle value).

False alerts: the number per patient per month. Keep each patient fully in either training or testing. Do not mix their days across both.

Evidence to Start

From Studies support combining home signals. Their strength differs. The figures come from abstracts and summaries. Check the full papers before you rely on them. You can use signals outside this list.

Signal	What studies show	How strong	Source
Daily weight	Researchers compared 268 admitted patients with similar patients who were not admitted. Weight gain started at least a week before admission. A gain of 2 to 5 pounds made admission about twice as likely.	Moderate. Some admissions come without weight gain.	Chaudhry et al., Circulation 2007 (PMC)

Several vital signs plus a personal baseline	A chest patch recorded heart rate, breathing rate and activity in 100 patients after discharge. Software predicted a return to hospital. It caught 76 to 88% of cases. It stayed quiet on 85% of normal days. The alert came about 6.5 days early. The patch takes more effort than this challenge allows.	Moderate to high. Small study with only 2 women.	Stehlik et al., Circ Heart Fail 2020 (paper)

Impedance of the chest	Implanted devices show impedance falling about 2 weeks before admission. A wearable vest was tested in 91 patients. It caught events that weight alone missed. Many of its alerts were false.	Moderate. Versions without surgery are still research prototypes.	Yu et al., Circulation 2005 (PubMed); vest and weight study
Body vibration from each heartbeat	A wearable patch with machine learning told stable patients from worsening patients. It used their response to light exercise. Another study used vibration data from clinic and home. I have not reviewed its results.	Low to moderate. Small studies from one centre.	Inan et al., Circ Heart Fail 2018 (PMC); Aydemir et al., IEEE Trans Biomed Eng 2020 (author list)

Extra heart sound (S3)	S3 is an extra sound after the normal heartbeat. A computer detector found high heart filling pressure 41% of the time. It found weak pumping 52% of the time. It found a high blood marker of heart strain 32% of the time.	Low alone. Use it with other signals.	Marcus et al., JAMA 2005 (abstract)

Signal	What studies show	How strong	Source
Heart rate variability (HRV)	HRV is how much the time between heartbeats changes. Low HRV predicted death in long-term heart failure. It does not show a coming episode.	Moderate for the long-term outlook. Low for early warning.	Nolan et al., UK-HEART, Circulation 1998 (reference)

Evaluation Criteria

Criterion	Weight
Creativity and originality of the sensing approach	20
Ease of use and likely long-term use by patients	20
Software and analytics	20
Hardware design and simulation proof	15
Scientific basis and honest testing	15
Cost, fit for India and pitch	10
Total	**100**

Deliverables

User journey (one page or storyboard)

Block diagram, circuit diagram and simulation proof

Parts list with prices in INR

Public GitHub page with code and setup steps Working demo using computer-generated data

Report of up to 2 pages covering method, evidence, results and limits

5-minute pitch

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Rules, Ethics and Disclaimer:

This is a monitoring idea. It is not a medical diagnosis tool. Say this clearly in your submission. Do not claim more accuracy than your tests show. Use public or computer-generated data only. Collecting data from people requires written consent and approval from the ethics committee (which is not available for this hackathon).

Teams can have 3 to 5 members.

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AMD

AMD Problem Statements

NIT Trichy · 24-25 October 2026 · Hardware track partner: AMD

Confirmed by Dr. Jayashree HV, AMD University Program (AUP), on 23 October 2026. Both problem statements are to be built on AMD devices.

Problem Statement 1: On-device AI assistant

Build an assistant that runs fully on a Ryzen AI PC with no cloud calls, judged on speed, accuracy and privacy.

Problem Statement 2: Quantise and deploy

Take an open model, quantise it with AMD tools, and show the gain in speed and memory with little loss in accuracy.

Resources from AMD

- **Compute:** remote access to 100 AMD Ryzen AI PC devices for the hackathon,

through the AMD University Program cloud.

- **Access:** each student gets their own URL and token. The URL and token map 1:1, and each token gives the student a separate workspace. Members of the same team will share the same device. Tokens are personal and must not be shared. The organisers will send them out separately.
- **Onboarding:** Dr. Jayashree offered to show how to use the tokens (AMD Compute Access + ROCm Software Stack Introduction session, invite sent 22 Oct) and to test access with a couple of students on a call.
- **Prizes:** AMD plans to send AMD goodies for the winners. There is no cash prize.
- **Support:** Dr. Jayashree HV (Jayashree.HV@amd.com), via the organising team.

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EMBIEN TECHNOLOGIES



THE LINK:

Embien_Transfinitte26_Problem_Statements (1).pdf

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ENTREPRENEURSHIP

CEDI
NIT TRICHY

Cash reward of 10k upfront and 2Lakh as seed grant if chosen

Focus Area : AI for Social Good

Sustainable Development Goods (SDG's)

Under NAVIYA - Pre Incubation Program :

- **UPTO INR 2 LAKHS** (Reimbursable GRANT) to build **MINIMUM VIABLE PRODUCT (MVP)** will be supported for **Top 15-20 Potential Startup Ideas.**
- **Tech and Business Mentoring Support**
- **Co-working Infra & Lab Facilities**
- **Investor connect & Demo Day**

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NITT FIRE

Competition Problem Statement

“Design a practical, fundable, and implementable 24-month roadmap for FIRE–NIT Trichy’s proposed research and entrepreneurship ecosystem–so that NITT can become a credible hub for student, faculty, and alumni-led venture creation. Your proposal should be tailored to NITT’s actual strengths, constraints, and regional opportunities, rather than simply copying IIT Madras. Address governance, infrastructure, faculty incentives, alumni engagement, corporate partnerships, funding, founder support, branding, and measurable outcomes.”

Deliverable:

A concise 8–12 page strategy report, including:
• Diagnosis of NITT’s entrepreneurial strengths and gaps
• FIRE operating model and governance structure
• Founder journey from idea to incorporation, first customer, and follow-on funding
• Physical and digital infrastructure plan
• Funding strategy and capital ladder
• Faculty, alumni, management, and student engagement plan
• Brand identity, including a logo concept and rationale
• 24-month implementation roadmap with KPIs
• Risks, assumptions, and mitigation plan
• One-page pitch for potential investors, corporate partners, or government stakeholders

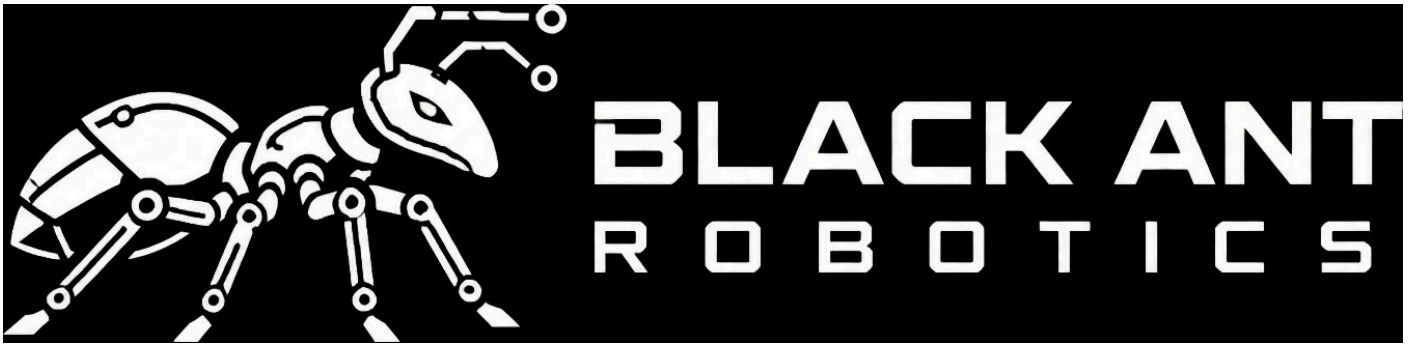
Evaluation focus:

Practicality, originality, understanding of NITT’s context, quality of the founder journey, funding realism, stakeholder engagement, and measurable impact.

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SOFTWARE

BLACK ANT ROBOTICS



1. The challenge in one page

Your team builds a computer simulation of a machine that tries to move Earth to a new orbit, for example from 1 AU out to about 2.7 AU near the asteroid belt, without harming the planet or life on it.

The reference concept: Octo-Earth

- A structural ring built around Earth along a great circle.
- Eight articulated arms rooted on the ring at equal 45° spacing, all at the same mounting radius.
- Arms up to about 12,000 km long, with segments, hinges, motors and a gripper at each tip.
- Conductors in the arms carry current that interacts with magnetic fields through the Lorentz force.
- The arms extend together on coordinated trajectories and keep the system balanced.

You may simulate Octo-Earth, change it, or propose another concept that acts on a planet-sized body: a gravity tractor, steered asteroid flybys, a mass driver, a solar sail or mirror array, a magnetic sail.

To win, your concept must work

Winning condition (all required)	What it means in practice
Earth reaches its target orbit	Within 10,000 simulated years
Earth stays safe	Every threshold in the rulebook appendix holds for the whole run
The structure survives	Every structural safety factor stays at or above 1.5
The code is validated	Every applicable benchmark passes
The physics is honest	Every force booked with its reaction; energy and momentum conserved
No unresolved flags	Only flags caused by your declared hypothetical enablers are allowed

You may declare up to three hypothetical enablers (section 7). Entries that honestly show their concept failing are still scored and can win other awards.

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UTHO

The Problem

Most cloud bills hold a quiet tax: money spent on capacity nobody uses. Teams size virtual machines for the worst day they can imagine and then never revisit them. Dev and staging servers run through nights and weekends. Disks and IPs outlive the projects that created them. By the time finance asks why the bill went up, nobody can explain where the money went.

The root cause is not carelessness. Teams can't easily see what they have, how much of it is actually used, and what it would be safe to change. Raw metrics exist, but turning them into decisions takes time and expertise that most teams don't have.

The Challenge

Build **SmartCloud Optimizer**, a working prototype that takes cloud infrastructure utilization data and turns it into decisions. It should show users where they are wasting resources, what they should change, how much they would save, and how risky each change is.

The goal is not just a dashboard of graphs. The goal is a tool an infrastructure engineer would actually trust enough to act on.

Scenario

Your users are the engineering team at a mid-sized Indian SaaS company. They run about 50 to 150 VMs across Noida, Bangalore and Mumbai, split across production, staging and development environments. Their monthly cloud bill has grown 40% in six months and leadership wants answers. Your platform is the first thing they open.

Data

You may generate your own simulated dataset or integrate with real cloud APIs

where feasible. Simulated data should be realistic and should include at least the following:

Inventory: resource ID, name, type (VM, block volume, public IP, etc.), region, environment tag (prod, staging or dev), owner or team, vCPU, RAM, disk size, and creation date.

Metrics: time series of at least 30 days at 5 to 15 minute granularity, covering CPU %, memory %, disk usage and IOPS, and network in/out.

Pricing: an hourly or monthly cost per resource or plan. Illustrative pricing is fine as long as it is clearly stated.

Strong submissions will deliberately include realistic patterns such as:

- steady low-usage VMs,
- bursty workloads with short daily peaks,
- batch jobs that are idle most of the day,
- dev environments idle on weekends,
- orphaned resources like unattached volumes and unused IPs,
- at least one genuine anomaly.

Core Features (Must-Have)

Resource Utilization Dashboard

Show CPU, memory, storage and network utilization across all resources. Users should be able to filter by region, environment and team, and drill down into any single resource's history.

Rightsizing and Cost Optimization Engine

Identify over-provisioned, underutilized and idle resources, and recommend a specific target configuration for each. Recommendations must be based on peak behaviour (for example p95 or p99), not just averages, so the tool never suggests shrinking a VM that would fall over during its daily peak.

Actionable Recommendations

Each recommendation should state the action, the reason, the estimated monthly saving, and a risk or confidence level. Actions should include:

- resize the instance,
- stop or delete an idle resource,
- clean up orphaned volumes and IPs,
- schedule non-production workloads to shut down off-hours.

Before vs After Cost Estimation

Compare the current monthly cost with the projected cost after applying the recommendations. Show the savings per recommendation, per environment and in total. Users should be able to accept or reject individual recommendations and

see the totals update.

Alerts and Insights

Flag unusual usage spikes, sudden cost increases, resources trending toward capacity limits, and projected budget overruns. Each alert should explain why it was raised.

Bonus Features (Good-to-Have)

Cost and usage forecasting for the next 30 to 90 days.

A "what-if" simulator, for example "what if we shut down all dev VMs from 8 PM to 8 AM?"

Plain-language explanations of recommendations, LLM-assisted or rule-based.

Export of recommendations as a report, or as scripts or IaC snippets.

Sustainability view showing estimated energy and carbon reduction from optimization.

Multi-tenant or role-based views (engineer vs finance vs leadership).

Expected Deliverables

Working prototype or MVP that runs locally or on free infrastructure.

Dashboard demonstrating utilization and optimization insights on your dataset.

Recommendation engine with sample cost-saving calculations.

Assumptions sheet listing pricing, thresholds, data generation logic and any simplifications.

Demo video or live demo (5 to 7 minutes) covering the problem, a walkthrough, the technical architecture, and real-world impact.

Source code repository with a README containing setup steps.

Evaluation Criteria

Criterion	Weight
Effectiveness and safety of optimization logic (do the recommendations make sense, and do they respect peak load?)	25%

Technical implementation and functionality	20%
Accuracy, clarity and transparency of cost estimates	20%
Innovation and originality	15%
Usability and dashboard design	10%
Scalability and practical applicability	10%

Rules and Notes

Participants are **not required** to purchase cloud services or use paid infrastructure. Simulated datasets and local environments are fully acceptable.

All cost-saving figures must state their assumptions clearly. **Unexplained savings numbers will be scored down**, however impressive they look.

Open-source libraries, frameworks and AI tools are permitted. Teams must be able to explain every part of their system.

Any technology stack, language, database or analytics tool may be used.

Objective

Build a practical tool that makes cloud infrastructure transparent, efficient and cost-effective. It should be something a real engineering team could open on Monday morning and act on before lunch.

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MCP to the moon



https://drive.google.com/file/d/1qQ9hfnzygtYTOZKRRr3m4FHfC1RvjU5M/view?usp=sharing

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MongoDB Student Hackathon - Challenge Tracks

Choose one track and build a working prototype.

Each track is open-ended. You can use public data, synthetic data, team-created material, third-party APIs, or data generated by your own product. Keep the input set small, make setup repeatable, and focus on completing one user workflow end to end.

Coding agents such as Codex or Claude Code are allowed. Be ready to explain your architecture, data model, and the choices you made.

Before the event

If you're new to building AI applications, we recommend AI Data Strategy with MongoDB as an introduction before the event.

Each track also includes relevant MongoDB Skill Badges for preparation or further learning afterwards. These are optional resources, and completing them is not required to participate.

Using Vector Search? You'll need embeddings, which represent your content as numerical vectors, and a Vector Search index configured to search them. MongoDB's automated embedding can simplify setup where supported. Check the linked guide for requirements. If you generate embeddings yourself, configure the index to match your model's output and use the same embedding model for your content and queries.

Track 1: Agentic Memory

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Build an application or agent that remembers useful information and changes how it behaves over time.

Challenge Brief

Build an application or agent with persistent memory across interactions or sessions. Decide what should be remembered, how it is retrieved, and when it should be updated, corrected, or forgotten. Represent memory that matters - facts, preferences, decisions, outcomes, constraints, feedback, or learned strategies. The stored memory must visibly change the system's next response, recommendation, plan, or action. The domain and interface are up to you.

Example Demo Flow

- Start with an interaction that creates a useful memory.
- Open a later session and show the retrieved memory changing the result or action.
- Inspect the stored memory and explain why it was selected.
- Demonstrate one difficult case: a correction, expiry, deletion, conflict, or private information.

Using MongoDB

- Model each memory as a document with an owner, type, content, source, timestamps, and lifecycle fields.
- Use MongoDB Vector Search for semantic recall, with filters to scope results by user, workspace, memory type or recency. See the setup note above for embedding and index requirements.
- Use a TTL index on a date field for time-based memory expiry. Handle other rules, such as deleting a memory when a flag changes, in application logic.
- Use change streams when multiple users or agents need live updates.

Relevant Docs

- Build AI Agents with MongoDB - Agent architecture, retrieval, and persistent memory.
- Data Modeling - Structure memory documents and their relationships.
- MongoDB Vector Search - Recall relevant memories with semantic search and metadata filters.
- TTL Indexes - Automatically expire time-limited memories.
- Change Streams - React to memory updates across users or agents.

Relevant MongoDB Skill Badges

- Memory for AI Applications
- Relational to Document Model
- Schema Design Optimization
- Vector Search Fundamentals

Example Project Ideas

- **Shared Context for Teammates' Agents** - separate coding agents share decisions, task ownership, file changes, constraints, and open questions.
- **Adaptive Incident Response Agent** - an agent remembers previous incidents, attempted fixes, and outcomes while detecting when old information no longer applies.
- **Long-Running Decision Assistant** - an assistant remembers assumptions, decisions, and results across a project and surfaces them when relevant.

Track 2: RAG Engine

Build a product that turns your own source material into grounded, useful results.

Challenge Brief

Build a RAG application around a problem of your choosing. Source a focused collection of documents, webpages, images, transcripts, or structured records - from a public source, your own material, or a set you generate with an LLM. Retrieve evidence from more than one source or section, then use it to answer, compare, recommend, investigate, or generate an artefact. Show the source behind every important result, and handle at least one difficult or unanswerable request clearly.

Example Demo Flow

- Show how a user adds or imports a small collection and how it becomes searchable.
- Run one request that requires evidence from multiple sources or distant sections.
- Open the citations and show the exact passages or source items used.
- Try a weak, conflicting, or unanswerable request and show how the product responds.

Using MongoDB

- Store original sources, chunks, embeddings, metadata, and source identifiers together. Use MongoDB Vector Search for semantic retrieval.
- Add metadata filters for dates, owners, or content types. Use hybrid search when exact names, codes, or keywords also matter.
- Keep a small set of test requests and expected source IDs so you can compare retrieval changes as you build.

Relevant Docs

- RAG with MongoDB - Build the ingestion, retrieval, and generation workflow.
- MongoDB Vector Search - Index embeddings and filter retrieval by source metadata.
- Hybrid Search Tutorial - Combine keyword and semantic search results.

- Explain Vector Search Results - Inspect query execution when debugging retrieval.

Relevant MongoDB Skill Badges

- Vector Search Fundamentals
- Vector Search Performance
- RAG with MongoDB
- Retrieval Evaluation

Example Project Ideas

- **Technical Due Diligence Assistant** - combine architecture documents, incident reports, and a repository summary into an evidence-backed engineering risk brief.
- **Policy Change Tracker** - compare document versions, identify changed obligations, and link each finding to the relevant passage.
- **Multimodal Repair Assistant** - combine manuals, error logs, and an inspection image to suggest next steps with cited instructions.

Track 3: Developer Tools for Production use cases

Build a tool that helps another developer understand, debug, or safely change a software or AI project.

Challenge Brief

Build a developer tool that improves one part of the build, test, debug, or release workflow. Pick one precise question or action - why a run failed, what changed, whether a release is safe, or which consumers will break. The tool can be a CLI, SDK, API, webhook, GitHub integration, editor extension, or a focused web interface. Run it against your own project or a generated sample, store the relevant history in MongoDB, and turn that history into a diagnosis, comparison, warning, or action. An LLM is optional.

Example Demo Flow

- Start by running the tool or triggering its integration against a real or generated project.
- Show the project activity or history captured in MongoDB.
- Run a second scenario with a meaningful change and show the result updating.
- Trigger one practical failure case: a duplicate event, incomplete input, failed job, or invalid configuration.

Using MongoDB

- Store runs and events with stable IDs, timestamps, and links to the project or version.

- Create indexes for the questions your tool asks most often, then use aggregation pipelines to compare runs or derive findings.
- Use schema validation for required event fields while keeping source-specific payloads flexible.
- Use change streams for live views or notifications, and make repeated ingestion safe with stable event IDs.

Relevant Docs

- Data Modeling - Design collections for runs, events, and project versions.
- Aggregation Operations - Compare runs and summarize event history.
- Schema Validation - Validate required fields in incoming events.
- Unique Indexes - Enforce unique event IDs to prevent duplicate records.
- Change Streams - Update live views when new events arrive.

Relevant MongoDB Skill Badges

- Relational to Document Model
- Fundamentals of Data Transformation
- Advanced Schema Patterns and Antipatterns

Example Project Ideas

- **Agent Run Debugger** - capture prompts, retrieved context, tool calls, outputs, and errors, then compare a successful run with a failed run.
- **API Change Impact Mapper** - compare two API specifications with a small consumer map, identify breaking changes, and generate migration tasks.
- **Coding Agent Change Verifier** - connect a task, agent plan, code diff, and test results, then flag requirements without evidence and risky changes without coverage.

General Hackathon Advice

- **Find your spark.** For inspiration, check out examples like Rajan Agarwal's projects, then put your own spin on an idea that excites you.
- **Try something new.** Build with MongoDB and use the day to explore an unfamiliar framework, API, or interface. A small working experiment is a great place to start.
- **See what models can really do.** Try a task you assumed was out of reach, test where the model succeeds or struggles, find its limits and build around what you discover.
- **Talk to the room.** Share what you're building, ask for help, and meet people whose skills complement yours. You might find a teammate or unlock a new direction.
- **Have fun with it.** Make something you're curious about, check out other teams' demos, and leave with new friends and something you learned.

ALL THE BEST TO ALL THE PARTICIPANTS